

Dimension Reduction Analysis: pathmnist

Generated report #2

Seed: 0 | Plan revision: 2

Dataset profile & preprocessing

PathMNIST, part of the MedMNIST v2 collection. Derived from a histopathology study of colorectal cancer: 28x28 RGB image patches extracted from hematoxylin and eosin (H&E) stained histology slides, downsampled from an original resolution of 224x224. Source: NCT-CRC-HE-100K (train/validation split) and CRC-VAL-HE-7K (test split, from a different clinical center; not loaded by this project). Only the train split is used.

Samples: 89,996 Features: 2,352 Sparsity: 0.0%

Missing values: No

Preprocessing: scale_unit_range

Modality: dense, naturally bounded-range (image/pixel-like). DATA_DESCRIPTION.md says the 2,352 features are raw 8-bit RGB pixel intensities, integers 0 to 255, and profile.json confirms value_range [0, 255] with 0.01% zeros (dense, unlike count data). The 0-255 bound is a storage encoding, not informative variation, so dividing by the fixed range maps every pixel to [0, 1] on the same scale. The bounds are fixed at 0 and 255 from the description rather than taken from the observed min/max, so the transform does not depend on which images happen to be in the split. Not standardize: every pixel shares the same native range, and standardizing would give low-variance pixels (for example near-constant background-colored positions) the same weight as informative ones. No normalize_total/loglp (these are not counts) and no select_top_variance: n_features = 2,352 is well below n_samples = 89,996, so dimensionality is not the bottleneck and dropping pixels would discard spatial information. Revision 2, judgment call: after running, PCA's first component explained 0.5465 of the variance against 0.0229 for the second, consistent with a per-image global intensity (brightness or stain strength) factor that this global rescale cannot remove. Per-image normalization was considered and rejected. The preprocessing vocabulary has no per-image centering step, and the only per-row step, normalize_total, rescales each image to a common pixel sum; in H&E patches overall lightness is itself part of what distinguishes tissue (pale background or fat versus densely stained nuclei), so removing it would discard likely real signal along with any stain-strength variation, and no available step separates the two. Consequence stated in the report: PC1 is not interpreted as tissue structure, since it may mix intensity or staining with tissue differences.

Methods and hyperparameters

pca (role: general_purpose, status: reused)

Hyperparameters: `n_components=50`

Required linear baseline and the only general-purpose method that scales comfortably to 89,996 images. `n_components = min(50, floor(89,996/2), 2,352) = 50`. Revision 2: the plan expected a few dozen components to capture most of the variance, because neighboring pixels and the three color channels of a pixel are strongly correlated. The observed spectrum is different: one dominant component (0.5465) followed by a slow tail, so 50 components retain 0.7907 of the variance and 178 are needed for 90%. Judgment call: 50 is kept as a compact baseline. Raising it toward 178 would mostly add fine pixel-level texture variance, and PCA's trustworthiness at 50 components is already 0.982 ($k = 10$, 5,000-image subsample), so local neighborhoods are well preserved at this size.

sparse_pca (role: general_purpose, status: reused)

Hyperparameters: `n_components=20, random_state=0`

The required PCA variant. With `n_samples = 89,996`, well above the ~5,000 threshold, Kernel PCA's 89,996 x 89,996 kernel matrix (about 65 GB in float64) is not feasible, so Sparse PCA is the variant that runs on the full data without subsampling. Its sparse loadings show which pixel positions and channels drive each component, which suits an image domain. Judgment call on `n_components = 20` rather than PCA's 50: method-selection/SKILL.md measured about 23 minutes for 50 components at this exact size, and cost grows with component count. 20 components keeps the run practical and still covers the leading directions. Alternatives: 50 (like-for-like with PCA but much slower), or Kernel PCA on a subsample (would not use the full data). Its variance explained is subspace-based (QR-orthogonalized), so it is not directly comparable per component with PCA's. Revision 2: the sparsity penalty is sklearn's default `alpha = 1`. `scripts/reduce_dim.py` does not pass an `alpha`, so it cannot be set from the plan; adding it to the hyperparameters would record a value the run never used. `Alpha = 1` on [0, 1]-scaled pixels gave components that together retain 0.7154 of the variance in 20 components, close to PCA's first 20, so the penalty did not shrink the captured subspace much; a larger `alpha` would give sparser, more localized pixel loadings at the cost of variance captured.

Methods and hyperparameters (cont.)

umap (role: visualization_only, status: reused)

Hyperparameters: `n_components=2`, `n_neighbors=30`, `min_dist=0.1`, `random_state=0`

Visualization only, never an input to any other computation. UMAP scales to 89,996 samples where MDS, Isomap and Diffusion Maps ($N \times N$ matrices) do not. `n_neighbors = 30` rather than 15 because `n_samples = 89,996` exceeds 20,000: a larger neighborhood keeps the graph connected and preserves more global arrangement at this size, while 30 is still about 0.03% of the samples, so it stays local. Judgment call on `min_dist = 0.1`: with 89,996 points in one plot, a small `min_dist` keeps dense regions compact so that any low-density gaps between them remain visible instead of being filled in; the alternative, a larger `min_dist` such as 0.5, would spread the main body out and make overplotting less severe but blur those gaps. DATA_DESCRIPTION.md calls raw pixel space a nonlinear image domain, which is what a neighbor-graph layout is suited to picture.

tsne (role: visualization_only, status: reused)

Hyperparameters: `n_components=2`, `perplexity=30`, `random_state=0`

A second visualization-only layout, so that any structural claim about the pictures has two independent methods behind it (CLAUDE.md guard rail; artifacts are keyed by method name, so a second UMAP run cannot be kept alongside the first). Barnes-Hut t-SNE (sklearn's default) is $O(N \log N)$, so it runs on all 89,996 images without subsampling, though it is the slowest step after Sparse PCA. `perplexity = 30` is an effective neighborhood of about 30 images, matched to UMAP's `n_neighbors = 30` so both layouts target the same local scale. Judgment call: a larger perplexity would show more global arrangement at a much higher cost at this N .

Quantitative metrics

method	trustworthiness	variance explained (PCs used)	PCs for 90% variance	label silhouette
pca	0.9820	0.7907	178	-0.0662
sparse_pca	0.9435	0.7154	>20	-0.0533
umap	0.8127			-0.0604
tsne	0.8347			-0.0336

How to read the metrics

Trustworthiness (`sklearn.manifold.trustworthiness`, 10 neighbours)

For each sample, takes its 10 nearest neighbours in the embedding and penalizes any that were not close neighbours in the original preprocessed data, weighting each penalty by how far down the original neighbour ranking that point sat. Ranges 0 to 1; 1 means the embedding introduces no false neighbours. It measures local neighbourhood fidelity only, not preservation of global distances. Computed against the data after the plan's preprocessing steps; on a random 5,000-sample subset when the dataset is larger than that (see Limitations).

Variance explained (PCs used)

The fraction of the total variance in the preprocessed data captured by the components actually kept in the embedding. For PCA, the sum of those components' explained-variance ratios (each component's variance divided by the total). For Sparse PCA, whose components are not orthogonal, no such per-component decomposition exists; instead this is the variance captured by the subspace the components span (the components orthogonalized via QR, then variance summed along that orthonormal basis) – a valid but different construction, marked in the sidecar JSON. Ranges 0 to 1. Blank for methods neither applies to.

PCs for 90% variance

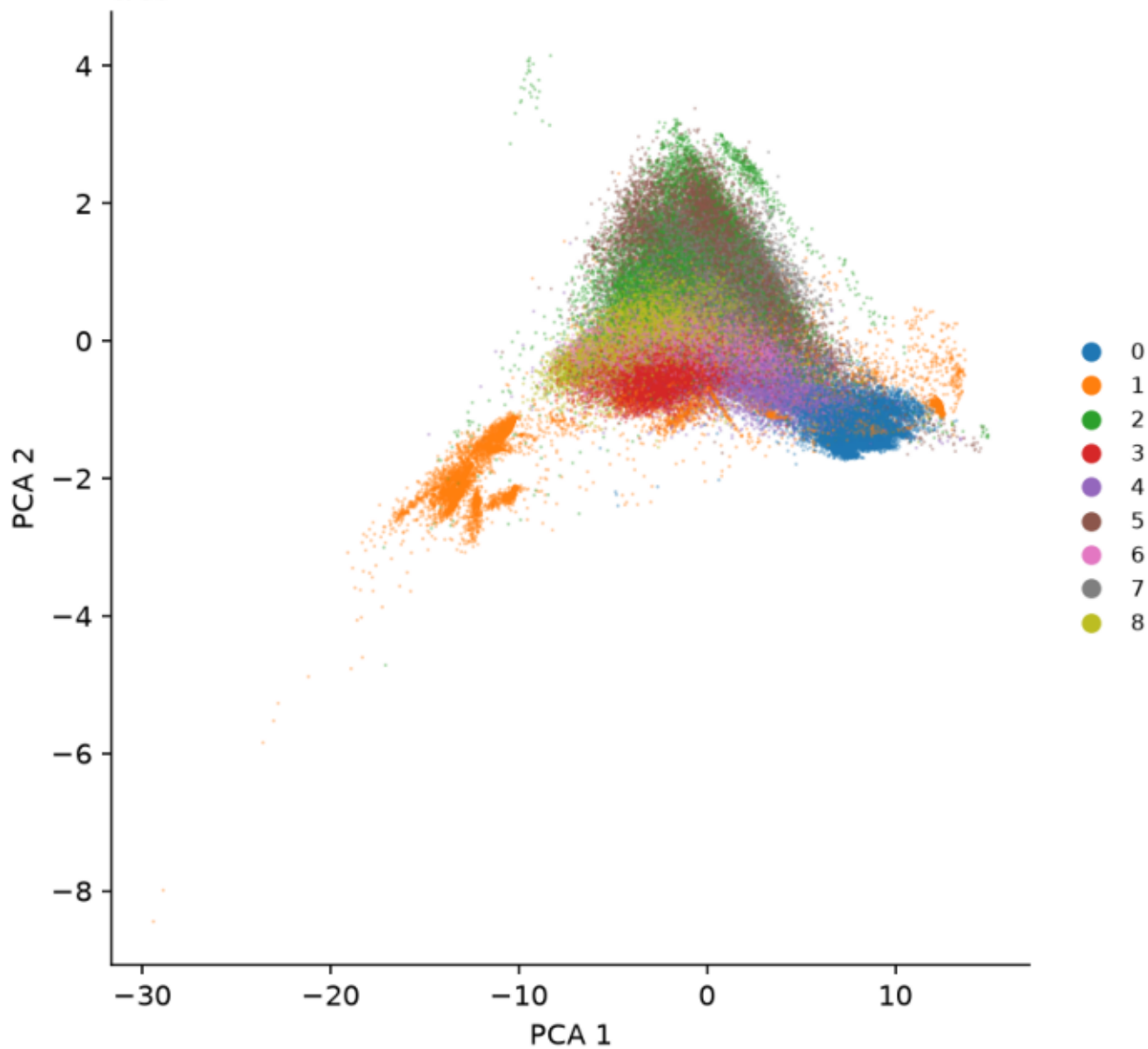
The smallest number of components whose cumulative explained variance reaches 90%. For PCA, read off a diagnostic fit of up to $\min(4 \times n_{\text{components}}, 200)$ components, deeper than what the embedding uses, so this can exceed the PCs actually kept. For Sparse PCA there is no deeper fit (it is too expensive to refit just for diagnostics); this is bounded by the components actually requested. Either way, a leading '>' means 90% was not reached within what was fit, so the true number is larger than shown. The scree plots show the full curve.

Label silhouette

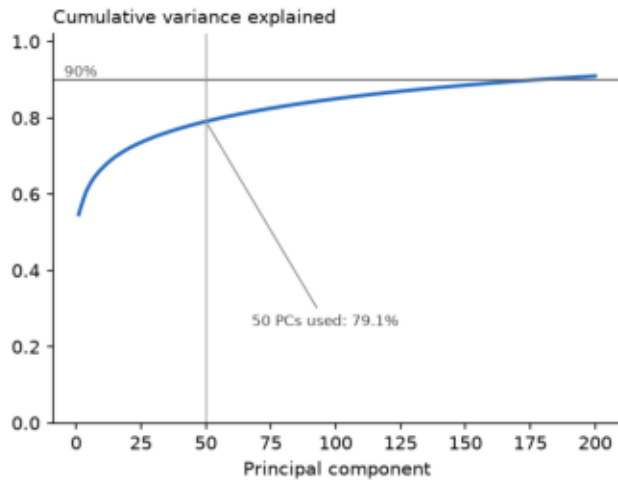
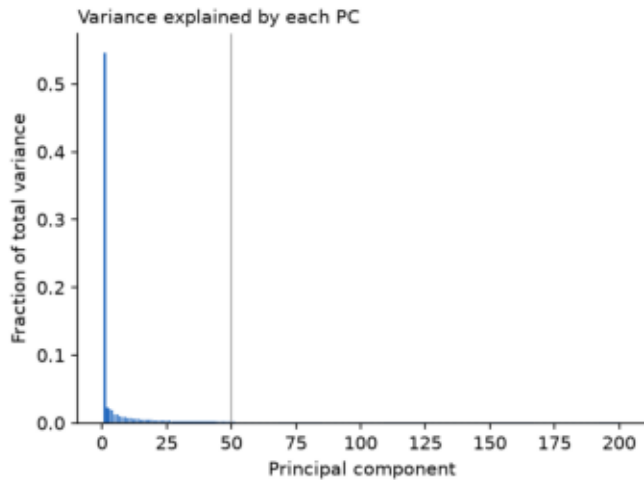
Mean silhouette coefficient (`sklearn.metrics.silhouette_score`) of the dataset's ground-truth class labels in the embedding space, using all embedding dimensions. Ranges -1 to 1: near 1 means points sit close to their own class and far from others, near 0 means classes overlap, negative means points are on average nearer another class than their own. A supervised sanity check only; labels were never used to build an embedding or to tune any hyperparameter.

pathmnist: pca

n=89,996, colored by ground-truth labels, showing dims 1-2
of 50

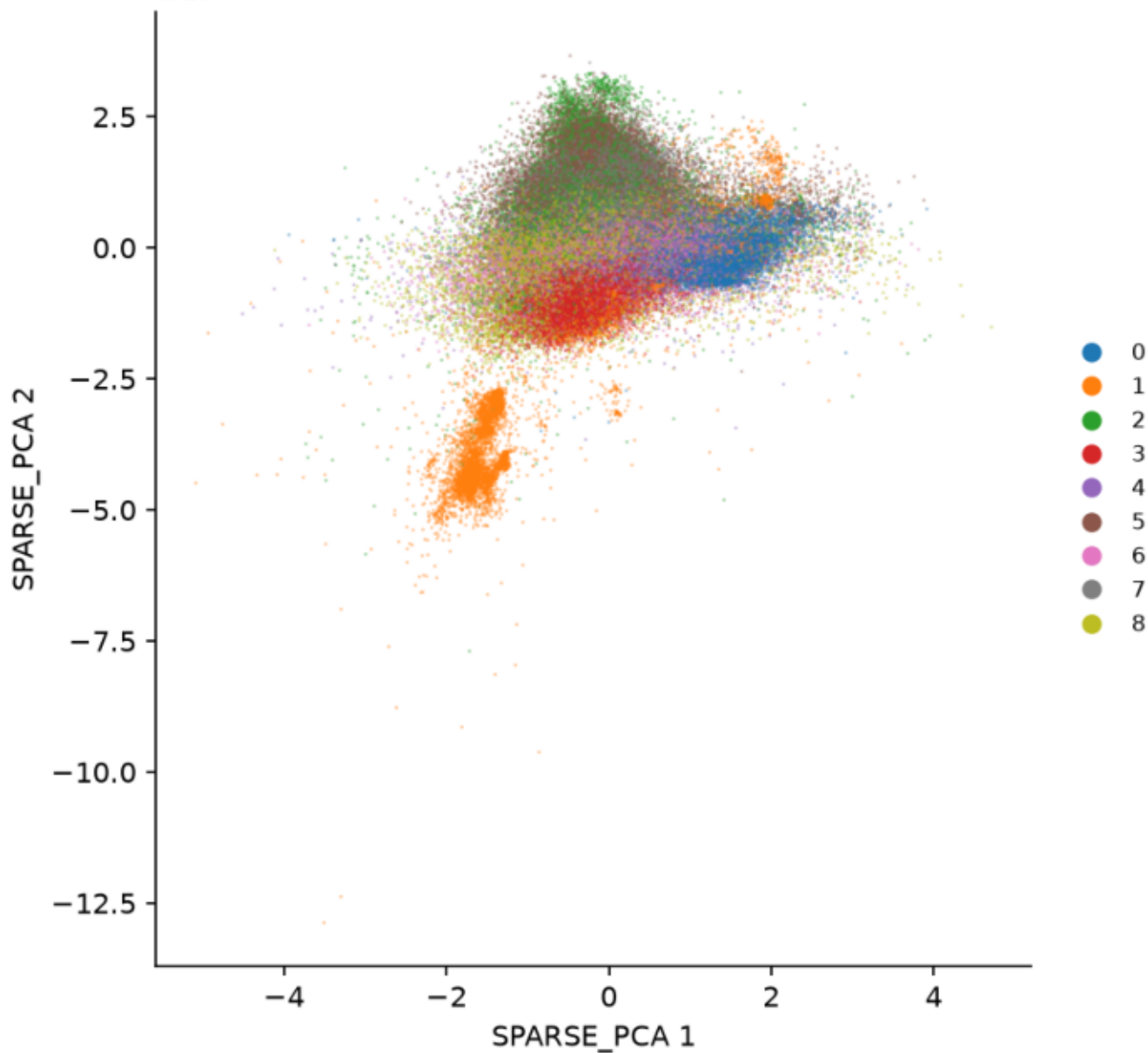


pathmnist: pca scree plot

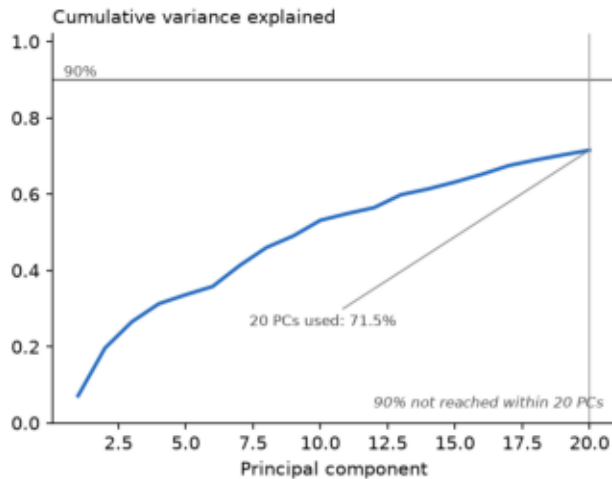
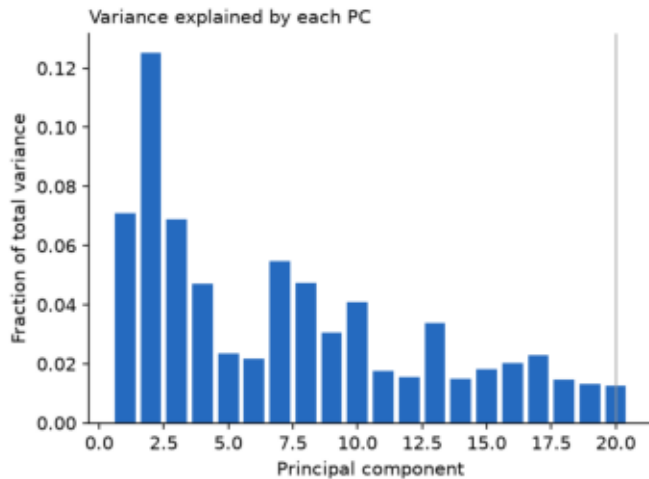


pathmnist: sparse_pca

n=89,996, colored by ground-truth labels, showing dims 1-2
of 20

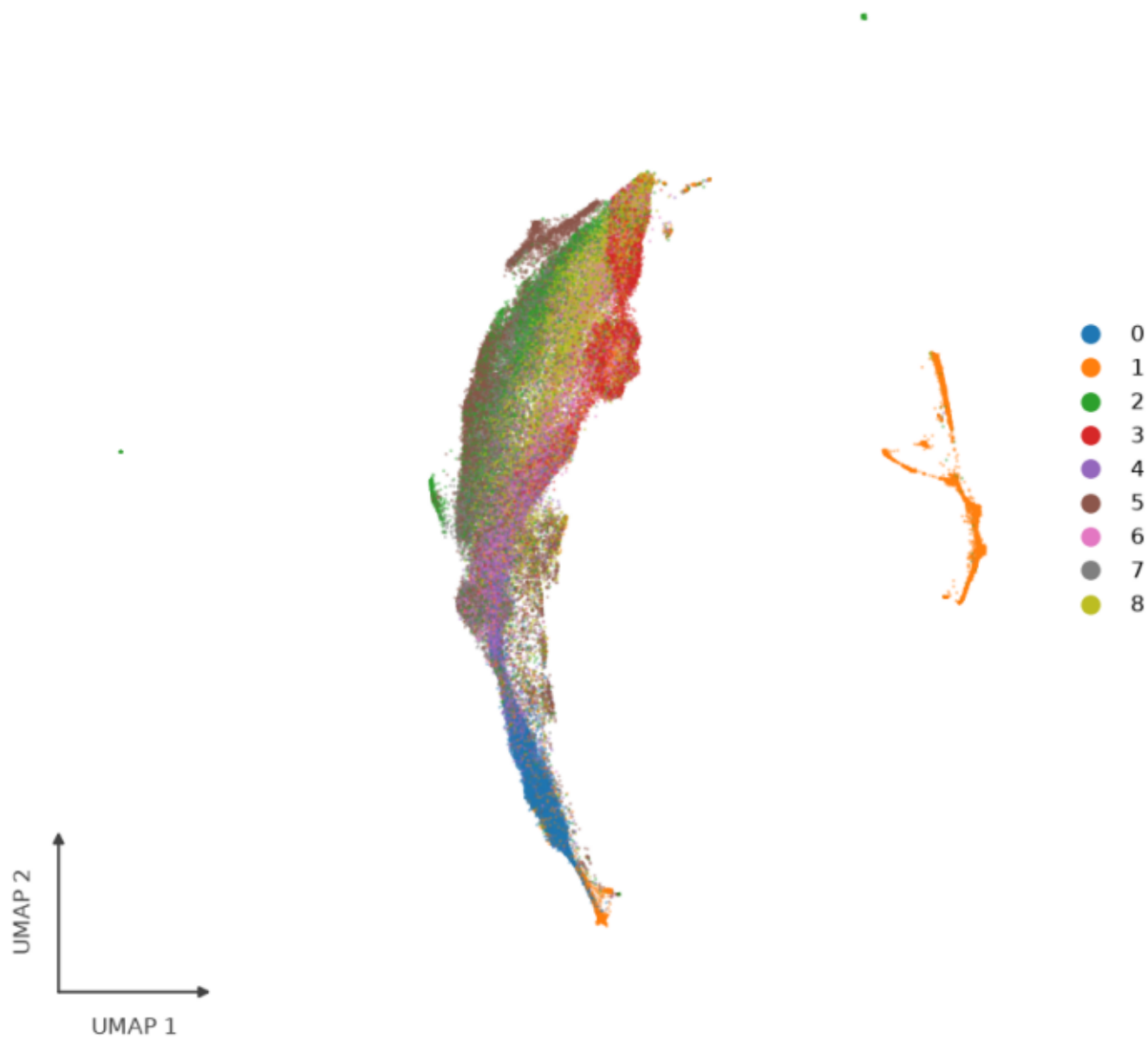


pathmnist: sparse_pca scree plot



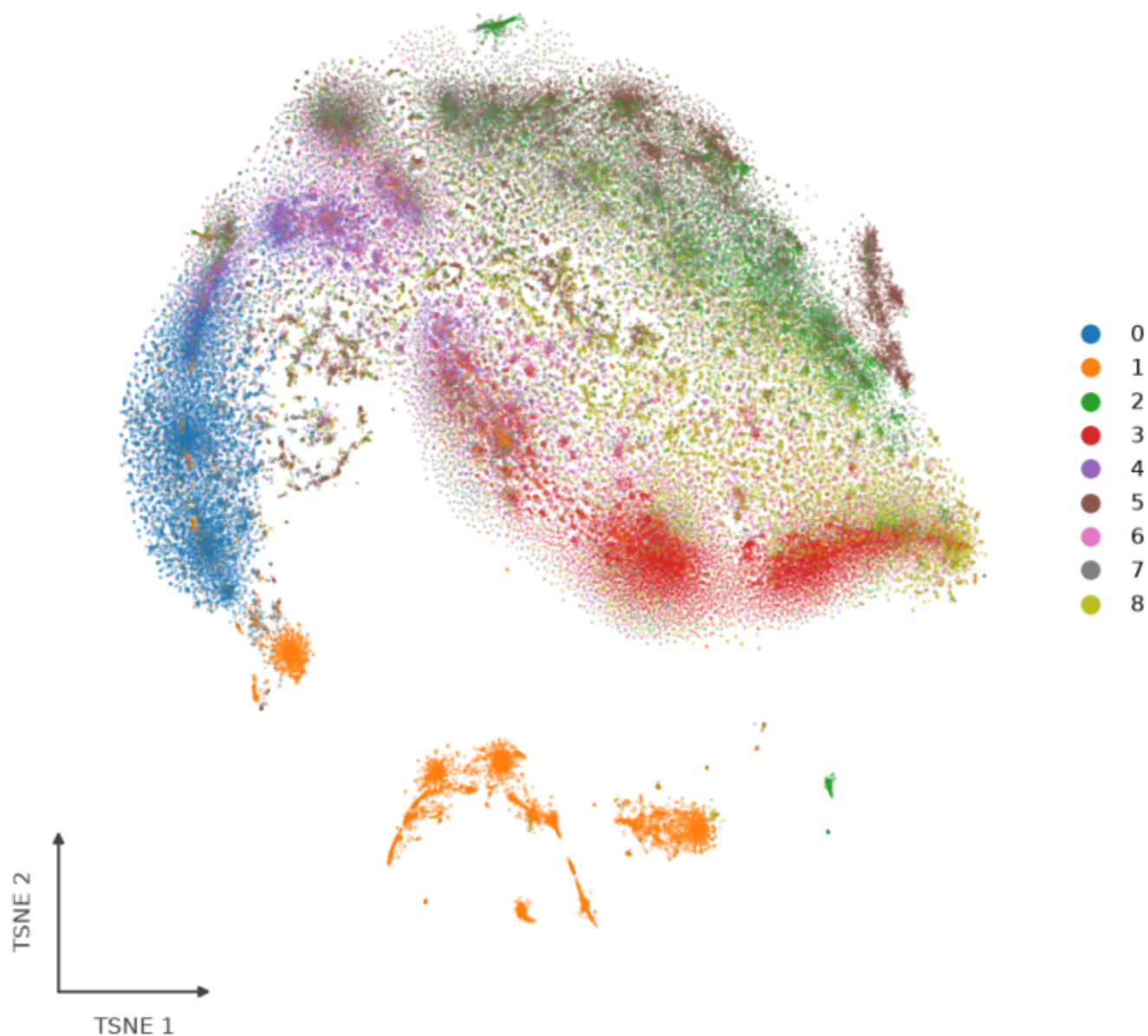
pathmnist: umap

n=89,996, colored by ground-truth labels



pathmnist: tsne

n=89,996, colored by ground-truth labels



Findings and interpretation

What each method produced

All four methods in the plan ran successfully on all 89,996 images, each flattened to 2,352 pixel values and rescaled from 0-255 to [0, 1]. Every stochastic step used seed 0. Trustworthiness and the label silhouette were computed on a fixed 5,000-image subsample, because the full data are too large for these $O(N^2)$ metrics.

PCA (50 components) retains 0.7907 of the total variance, and 178 components are needed to reach 90%. The spectrum is dominated by one direction: the first component explains 0.5465 and the next two explain 0.0229 and 0.0209. After that, the variance decays slowly. Trustworthiness in 50 dimensions is 0.9820.

Sparse PCA (20 components, sklearn's default sparsity penalty) retains 0.7154 of the variance in the subspace its components span. 90% is not reached within 20 components (>20). Its per-component values are not ordered by size the way PCA's are (its second component, 0.1253, is larger than its first, 0.0710), because Sparse PCA does not order components by variance. Trustworthiness in 20 dimensions is 0.9435.

UMAP (n_neighbors = 30, min_dist = 0.1) has trustworthiness 0.8127 and t-SNE (perplexity = 30) has 0.8347, both in 2 dimensions.

These trustworthiness values come from embeddings of different dimensionality (50, 20, 2 and 2). They do not rank the methods.

Label sanity check

The dataset has 9 pathologist-assigned tissue labels. They were used only to color the plots and to compute this check, never as input. DATA_DESCRIPTION.md lists the tissue names but not which integer maps to which name, so the plots and this text refer to labels by integer (0 to 8).

The label silhouette is negative in every embedding: PCA -0.0662, Sparse PCA -0.0533, UMAP -0.0604, t-SNE -0.0336. By this measure, images of the same label are on average no closer to each other than to images of other labels in any of these embeddings. That describes these unsupervised embeddings of raw pixels. It does not mean the tissue types are indistinguishable: silhouette is a global, distance-based score, and it is pulled down by the large overlapping central mass seen in every figure.

Findings and interpretation (cont.)

What the figures show

The first component is much larger than the rest (0.5465 against 0.0229). A factor that strong in raw pixel space is consistent with overall image lightness, which can mix staining strength with real tissue differences. For that reason, this report does not read PC1 as tissue structure (see the critique log).

A few things are consistent across all four embeddings. Most label-1 images sit apart from the main body of points: as a separate arm in PCA and Sparse PCA, and as separate fragments in both UMAP and t-SNE. Label-0 images form a fairly coherent region toward one side of the main body in all four. The remaining labels (2 to 8) overlap heavily in one central mass in PCA and Sparse PCA. In the two layouts, some of them occupy loosely preferred areas (for example, label 3 in part of the t-SNE ring and at the upper end of the UMAP body), but with a lot of mixing.

Beyond that, the two layouts disagree. UMAP shows one long connected body with label 0 at one end. t-SNE shows a ring with a large, mixed center. Because they disagree, and each was run with one seed, no claim is made about the shape, ordering or gaps within the main body. As with any UMAP or t-SNE picture, the sizes of and distances between groups are not measurements.

How the methods relate

PCA and Sparse PCA are the general-purpose representations. They give similar views: both separate much of label 1 and put label 0 at one side, and their first 20 components capture a similar share of variance. Sparse PCA was chosen over Kernel PCA because a kernel matrix at this sample size is not feasible. It was the slowest step after t-SNE and added little new structure in its first two dimensions. UMAP and t-SNE are pictures only. Their use here is to check which coarse patterns hold up under two different nonlinear layouts.

Findings and interpretation (cont.)

What can't be concluded

Flattened 28x28 pixels do not encode texture in a way that nearby vectors mean similar tissue (DATA_DESCRIPTION.md calls this a nonlinear image domain), and the negative label silhouettes show that none of these unsupervised methods separates most tissue labels in raw pixel space. A feature extractor designed for images (for example, a convolutional network) might, but that was outside this analysis. The dominant first component may partly reflect staining or lightness differences between images and slides rather than tissue type, and the available preprocessing cannot separate the two. Only the training split was analyzed, so nothing here speaks to the separate test split from another clinical center. All metrics are based on a 5,000-image subsample, and each manifold method was run with one seed.

Critique and revision

Reviewed plan revision 1. Readiness: revise: at least one finding was addressed with a plan revision.

5 finding(s): critical=0, moderate=2, minor=3

F1: preprocessing_mismatch (moderate)

PCA's first component alone explains 54.6% of variance, more than 20 times the second component, which is consistent with a per-image global intensity (brightness or stain strength) factor that the only preprocessing step, a fixed 0-255 to [0, 1] rescale, does not address.

Evidence: `metrics/pca.json` `explained_variance_ratio[0] = 0.5465`, `[1] = 0.0229`, `[2] = 0.0209`. `plan.json` preprocessing has exactly one step, `scale_unit_range` with params `{min: 0, max: 255}`, which is a global linear rescale and cannot change the relative variance of components. The structure check passes: there is one shared preprocessing list and no per-method preprocessing. In `figures/pca.png`, PC1 runs from about -30 to +15 and mostly spreads class 1 and class 0 apart, while the remaining classes overlap in a narrow band, so PC1 may carry real signal as well as an intensity confound. The metrics alone cannot tell these apart.

Recommendation: Either add a per-image intensity normalization step to the shared preprocessing block (for example, subtract each image's mean pixel value, or standardize each image), rerun, and compare the spectrum and label silhouette to this run. Or keep the current preprocessing but change the `scale_unit_range` reason to say that a dominant PC1 (54.6%) was observed, that it may reflect per-image brightness or staining rather than tissue structure, and that per-image normalization was considered and rejected, with the reason why. In both cases, the report should not interpret PC1 as tissue structure without this caveat.

Critique and revision (cont.)

F2: instability_or_metric (moderate)

The PCA reason expected that 'a few dozen components should capture most of the variance', but the metrics show 50 components capture 79.1% and 178 are needed to reach 90%, so the plan's stated expectation for `n_components = 50` does not hold.

Evidence: `plan.json` `pca` reason: 'a few dozen components should capture most of the variance; the scree plot and 90%-variance count will show whether 50 is enough.'

`metrics/pca.json`: `variance_explained_by_used_components = 0.7907`,
`components_needed_for_90pct_variance = 178` (not a lower bound, diagnostic fit to 200 components). After PC1, the spectrum decays slowly (`PC10 = 0.0069`, `PC50 = 0.0017`).

Recommendation: Rewrite the PCA reason to match the observed spectrum: 50 components retain 79.1%, and 90% needs 178. State whether 50 is still kept on purpose (for example, as a compact baseline, with the trade-off accepted) or raise `n_components` toward 178. If F1 leads to per-image normalization, re-read the spectrum after that change before choosing a number.

F3: unjustified_hyperparameter (minor)

Sparse PCA's sparsity penalty (`alpha`), its most important hyperparameter, is not set in the plan, so an implicit library default is used with no reason given.

Evidence: `plan.json` `sparse_pca` `hyperparameters = {"n_components": 20, "random_state": 0}`; no `alpha`. The reason explains `n_components` and the choice of Sparse PCA over Kernel PCA, and says 'its sparse loadings show which pixel positions and channels drive each component', but how sparse those loadings are depends entirely on `alpha`, which the reason never mentions.

Recommendation: Set `alpha` explicitly in `hyperparameters` and give a reason for it (or state that the default was kept as a judgment call, name the alternatives, and say why), in line with CLAUDE.md rules 11 and 12.

Critique and revision (cont.)

F4: unjustified_hyperparameter (minor)

The UMAP `min_dist = 0.1` reason is generic and would read the same for any dataset.

Evidence: `plan.json` `umap` reason: `'min_dist = 0.1 keeps dense regions compact, so gaps between groups stay visible.'` This cites nothing specific to `pathmnist`. By contrast, the `n_neighbors` part of the same reason does cite `n_samples = 89,996`.

Recommendation: Rewrite the `min_dist` clause to name something about this dataset that motivates it, or mark it as a judgment call and name the alternative considered (for example, a larger `min_dist` to spread out the dense main body).

F5: label_leakage_in_reasoning (minor)

The UMAP reason uses a label-derived number (the smallest class size, 7,886 images) as part of its justification for `n_neighbors`, even though it calls this only a scale check.

Evidence: `plan.json` `umap` reason: `'30 is still far below the smallest class (7,886 images per DATA_DESCRIPTION.md; used only as a scale check, not to tune toward the labels).'` The main justification (`n_samples = 89,996 > 20,000`) does not need label information, and the value 30 would be the same without this clause. So this is a leak in the reasoning, not in the choice itself, but it is still label knowledge in an unsupervised hyperparameter reason.

Recommendation: Remove the smallest-class clause from the UMAP reason. If an upper-bound sanity check on `n_neighbors` is wanted, state it in terms of `n_samples` only.

Critique and revision (cont.)

Critic's notes

Manifold overread check: this passes. There are two independent visualization_only methods, UMAP (n_neighbors 30) and t-SNE (perplexity 30), each run with a single seed (0). The structure the two layouts share is limited. Both put most of class 1 into separate fragments away from the main mass, and PCA and Sparse PCA also separate much of class 1 along their first two dimensions. Class 0 forms a fairly coherent region in both layouts. Beyond that they disagree: UMAP shows one long connected body with class 0 at one end, while t-SNE shows a ring with a large mixed center. So only claims both layouts support (class 1 separates, class 0 is fairly coherent, the other classes overlap heavily) should be made, not claims about shape, ordering, or gaps within the main body. Label silhouettes are negative for every method (PCA -0.0662, Sparse PCA -0.0533, UMAP -0.0604, t-SNE -0.0336, all on 5,000-sample subsamples). This fits the heavy overlap seen in the figures, and no plan reason explicitly predicted positive separation, so it is not listed as a finding. The report should still state it plainly. Trustworthiness values are computed in each embedding's own dimensionality (PCA 50-D: 0.982, Sparse PCA 20-D: 0.9435, UMAP 2-D: 0.8127, t-SNE 2-D: 0.8347, all k = 10 on 5,000 samples), so they are not a like-for-like ranking between methods, and the report should not present them as one. The PCA vs Sparse PCA comparison looks consistent: PCA's first 20 components sum to about 0.719, close to Sparse PCA's subspace-based 0.7154 at 20 components.

Revision applied (now revision 2)

dr-critic (revision 1) raised 5 findings; all addressed by rewriting reasons, no hyperparameter changed, so no method was rerun. F1: the dominant PC1 (0.5465) is acknowledged as possibly a per-image intensity or staining factor; per-image normalization was considered and rejected (no such step exists, and normalize_total would also remove lightness differences that likely separate tissues), and the report does not interpret PC1 as tissue structure. F2: PCA reason rewritten to match the observed spectrum (0.7907 at 50 components, 178 for 90%), keeping 50 as a stated judgment call. F3: Sparse PCA alpha documented as sklearn's default of 1, fixed by the implementation. F4: UMAP min_dist reason made dataset-specific. F5: label-derived class-size clause removed from the UMAP reason.

Limitations

Trustworthiness was computed on a random subset of 5,000 of 89,996 samples (it is $O(N^2)$ and infeasible at full size) for: `pca`, `sparse_pca`, `umap`, `tsne`. Treat those scores as estimates, not exact full-dataset values.

The label silhouette column is a supervised sanity check only, computed on the embedding's own dimensions; it was never used to tune any hyperparameter.

The label silhouette was scored on a fixed-seed random subset of 5,000 of 89,996 samples (it is $O(N^2)$) for: `pca`, `sparse_pca`, `umap`, `tsne`. Treat it as an estimate, not an exact full-dataset value.

`sparse_pca`: 'variance explained' and 'PCs for 90%' come from the variance captured by the subspace the (non-orthogonal) components span, not a per-component eigendecomposition like PCA's; see the metric definitions above. Also bounded by the components actually requested, not a deeper diagnostic fit, since refitting deeper is too expensive for this method at this dataset's size.